

# R<sup>2</sup>-Transfer: Reflective Retrieval-Augmented Multi-Agent Framework for Cross-City Time Series Prediction

Paper #307

## Abstract

Spatio-temporal prediction is central to urban computing tasks such as traffic management and urban planning, yet many cities lack sufficient data to train an accurate model. Cross-city knowledge transfer alleviates this problem by reusing knowledge from data-rich cities to support prediction in data-scarce ones. However, existing methods learn source knowledge globally; the correspondence that supports a given target prediction is often local and query-dependent, so much of what is transferred is irrelevant to a specific query. In this work, we propose R<sup>2</sup>-Transfer, a reflective retrieval-augmented multi-agent framework. R<sup>2</sup>-Transfer organizes source data into local entries, then retrieves the entries relevant to each target query, and predicts by analogy over their reference futures. When a large error is observed, a Reflection Agent traces it back to the responsible agent and revises the corresponding prompt or memory, allowing the workflow to improve without retraining. Extensive experiments on three source-target city pairs show that R<sup>2</sup>-Transfer achieves the best results on 15 of 18 evaluations, with mean MAE and RMSE improvements of 5.9% and 7.1% over the leading baseline. These results suggest that retrieval over local entry is a practical path toward accurate, self-improving prediction for data-scarce cities. Our source code is available at: <https://anonymous.4open.science/r/anonym261B>.

## CCS Concepts

• Information systems → Spatial-temporal systems; Data mining.

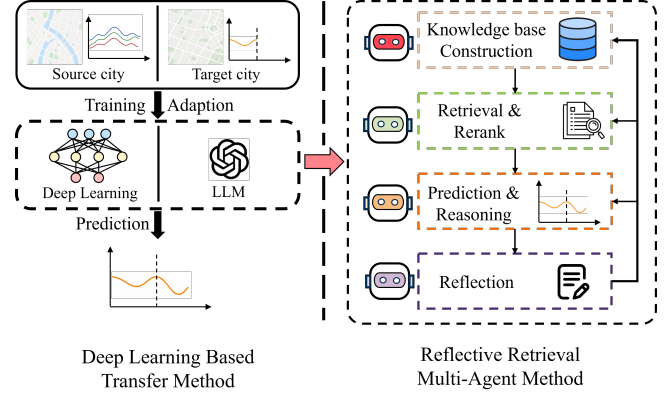
## Keywords

Cross-city prediction, Transfer learning, Multi-agent, Reflection, Retrieval-augmented.

## 1 Introduction

Spatio-temporal prediction supports a wide range of urban applications, including traffic control and urban planning. In recent years, deep learning-based spatio-temporal models [16, 28, 38, 42] have achieved remarkable predictive accuracy. However, strong predictive performance often depends on abundant training data. Many cities lack large-scale datasets because of limited sensing infrastructure, high data collection costs, or privacy restrictions. Insufficient training data could lead to performance degradation in real-world deployments. Cross-city transfer learning offers a promising solution by leveraging knowledge of data-rich city to improve prediction performance in data-scarce city. Specifically, cross-city transfer learning transfers useful representations, knowledge, or model parameters learned in the source domain to aid the target domain. Therefore, cross-city transfer learning can enable accurate few-shot prediction for city with limited data.

Existing work has explored this direction through deep learning methods [3, 12, 13, 21, 23, 44], transfer learning methods [35, 40],



**Figure 1: Comparison between conventional methods and our reflective retrieval multi-agent framework, where specialized agents collaborate on knowledge base construction, retrieval, prediction, and reflection.**

and LLM-enhanced methods [10, 11, 18]. Existing methods have made initial contributions to cross-city transfer learning, but they still face two main limitations. First, existing methods transfer the knowledge of a data-rich city globally. Global transfer typically compresses the entire knowledge of a data-rich city into dataset-level carriers, such as model parameters, shared latent representations, or a bank of pattern prototypes. However, the knowledge supporting a specific prediction task in a data-scarce city is often local and query-dependent. Different prediction tasks correspond to different spatio-temporal patterns. Knowledge with similar temporal pattern may come from locations with different spatial pattern. Differences in spatial pattern could lead to divergent subsequent evolutions of the time series. For example, a commercial center and a residential area may exhibit similar temporal patterns during the morning peak. After the peak, the traffic of the commercial center remains at a high level, whereas the traffic of the residential area drops rapidly. Therefore, only local knowledge whose temporal and spatial pattern both match the current prediction task can provide valuable references for transfer. In contrast, global carriers inevitably mix in a large amount of knowledge irrelevant to the current prediction task. Irrelevant knowledge is transferred as noise and degrades prediction performance. Second, existing methods implement the entire transfer as a single monolithic model trained end to end. In a monolithic model, all steps are entangled within shared parameters. As a result, a prediction failure cannot be attributed to the step that caused it. A large error may stem from missing knowledge, from selecting irrelevant knowledge, or from misusing relevant knowledge. However, the training loss only reflects the magnitude of the error, not the step in which the error occurred. Without step-level attribution, the erroneous step cannot be localized and corrected in isolation. The only remedy is to retrain the entire system, which is costly and cannot guarantee that similar errors will not recur.

To address these limitations, we aim to investigate a reflective retrieval-augmented framework for cross-city few-shot prediction. First, the framework preserves the knowledge of the data-rich city as a local knowledge base rather than compressing the knowledge into a global carrier. For each prediction task, only a few relevant knowledge is retrieved and transferred. So irrelevant knowledge is excluded from transfer. Second, the framework decomposes the transfer process into specialized agents. This modular design makes every step individually inspectable. Thus, a prediction failure can be traced back to the responsible agent. As a result, the agents in the framework keep improving after deployment, without retraining the entire system.

Achieving this framework requires addressing the following three main challenges.

- **How can the transferable knowledge base be extracted and maintained from data-rich cities?** The knowledge that is useful for a prediction is local rather than global. A transferable unit should therefore preserve the time series, and the corresponding spatio-temporal context. However, different datasets adopt different spatio-temporal patterns, so units are hard to transfer directly. The challenge is to convert raw data of units into aligned data and organize them into a high-quality, non-redundant knowledge base.

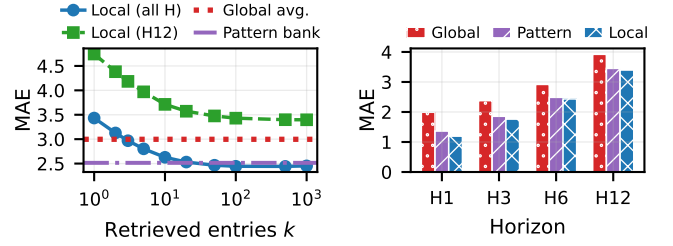
- **How can relevant local knowledge be retrieved for each prediction query?** Each query may match only a small subset of the knowledge base. Relevance of query is also multi-perspective. Temporally similar knowledge may be found in different spatial contexts, and spatially matched knowledge may follow different temporal patterns. The challenge is to follow spatial and temporal relevance to select the genuinely useful knowledge for each query.

- **How can prediction feedback drive reflection over the multi-agent framework?** A prediction failure may come from missing knowledge, mismatched retrieval, or biased prediction. Reflection should first attribute a cause of failure, then revise only the responsible agent. The challenge is to make such revisions effective while keeping the overall framework stable.

To address these challenges, we propose  $R^2$ -Transfer, a reflective retrieval-augmented multi-agent framework, as illustrated in Fig. 1. Specifically, the Knowledge Base Agent extracts local knowledge from data-rich city, aligns with knowledge of data-scarce city, and maintains a high-quality, non-redundant knowledge base. Given a prediction query, the Retrieval Agent first retrieves knowledge base by temporal similarity to form a candidate set. Then Retrieval Agent reranks the candidates from temporal and spatial perspectives to select the most relevant knowledge. The Prediction Agent produces a prediction and reasoning through prediction query data and the retrieved knowledge. When the prediction error exceeds the threshold, the Reflection Agent inspects the reason of failure, and updates the memory or prompt of the responsible agent. Each update is reverted if subsequent predictions do not improve. The four agents thus form a closed loop of knowledge construction, retrieval, forecasting, and feedback-driven adaptation, without any retraining.

In summary, the main contributions of this paper are as follows:

- We investigate cross-city few-shot spatio-temporal prediction under limited data. Useful cross-city correspondence is local and



**Figure 2: (a) Local Transfer vs. Global Transfer, which shows the analogical prediction error as a function of the number of source entries  $k$  used per query. (b) Local retrieval performances across horizons, which shows the horizon-wise MAE of Global-Average, Pattern-Bank, and Local-Retrieval.**

query-dependent, whereas existing methods mainly transfer holistic knowledge. We formulate the task as query-conditioned local knowledge transfer.

- We propose  $R^2$ -Transfer, a reflective retrieval-augmented multi-agent framework. The Knowledge Base Agent extracts and aligns knowledge and maintains a knowledge base. The Retrieval Agent selects relevant knowledge using spatial and temporal patterns. The Prediction Agent generates predictions and reasoning. The Reflection Agent diagnoses prediction failures and applies bounded memory or prompt revisions without parameter retraining.

- Extensive experiments across three cross-city few-shot settings show that  $R^2$ -Transfer achieves the best results on 15 of 18 evaluations.  $R^2$ -Transfer improves average MAE and RMSE by 5.9% and 7.1% over the leading baseline, respectively. Ablation studies and case analyses further demonstrate the benefits of aligned local knowledge, multi-perspective retrieval, and reflection-driven adaptation.

## 2 Preliminaries

### 2.1 Global Transfer vs. Local Retrieval

Source-city knowledge can be transferred in two paradigms. Global transfer compresses the entire source-city experience into dataset-level carriers. Local retrieval keeps source experience as explicit local entries and applies query-relevant entries to each target query. The choice hinges on whether the useful cross-city correspondence is globally shared or locally query-dependent. We conduct a model-free study on dataset pair M-P (METR-LA transfer to PEMS-BAY). We sample the source city into a pool of local entries, each storing a 12-step history and its 12-step future. For each target query, we forecast by analogy without any training on the target city. Three selection strategies are compared: Global-Average uses the mean future of the whole pool; Pattern-Bank uses the nearest global prototype after clustering; Local-Retrieval retrieves the  $k$  entries with the most similar histories. Fig. 2 yields two observations. First, the useful correspondence is local. The analogical error is minimized when only a bounded subset of nearest source entries is used, and rises toward the global extreme (Fig. 2a). Pooling the whole source city dilutes the informative knowledge. Second, local evidence helps at every horizon. Local-Retrieval outperforms Global-Average and Pattern-Bank across all horizons (Fig. 2b), reducing MAE from 1.99 to 1.20 at the first horizon and from 3.92 to 3.40 at the twelfth.

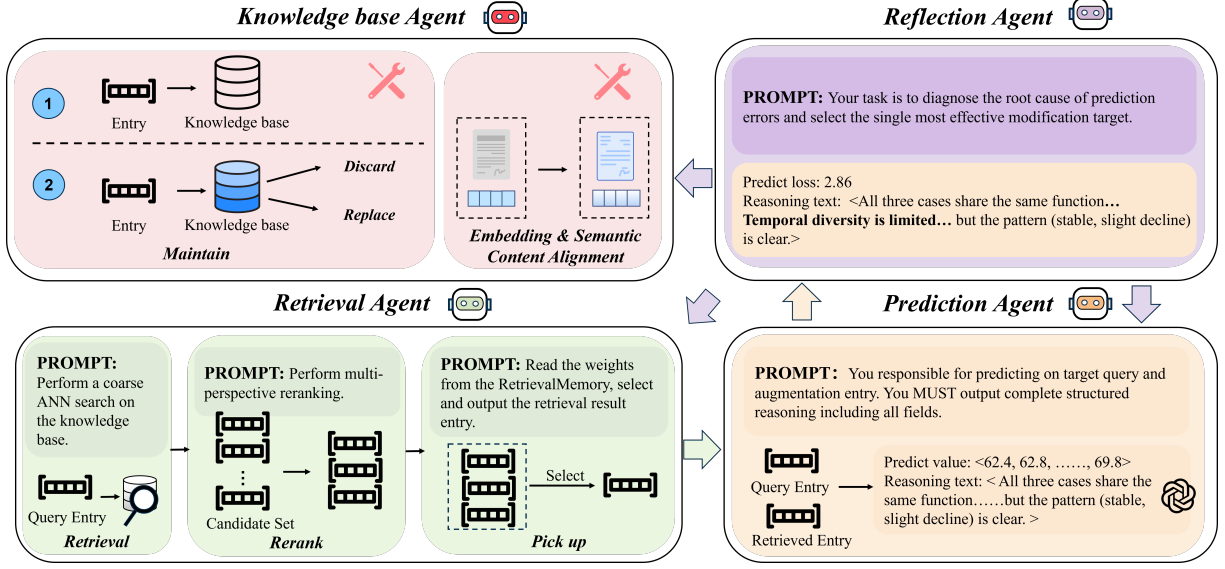


Figure 3: The overall architecture of R<sup>2</sup>-Transfer, which consists of four collaborative agents: a Knowledge Base Agent, a Retrieval Agent, a Prediction Agent and a Reflection Agent.

## 2.2 Problem Definition

Formally, a city is represented by a spatial structure  $\mathcal{S} = (\mathcal{V}, \mathbf{A})$ , where  $\mathcal{V}$  is a set of  $N = |\mathcal{V}|$  observation nodes and  $\mathbf{A} \in \mathbb{R}^{N \times N}$  is the adjacency matrix representing their connectivity. Each node records a signal at a fixed sampling interval  $T$ , yielding an observation matrix  $\mathbf{X} \in \mathbb{R}^{T \times N}$ , where  $\mathbf{X}_{t,v}$  denotes the value at node  $v$  and step  $t$ . Given an input window of length  $P$  and a prediction horizon of length  $Q$ , the standard prediction task predicts the future signals of all nodes from their recent past:

$$\hat{\mathbf{X}}_{t+1:t+Q} = \mathcal{F}(\mathbf{X}_{t-P+1:t}, \mathbf{A}), \quad (1)$$

where  $\mathbf{X}_{t-P+1:t} \in \mathbb{R}^{P \times N}$  is the input and  $\hat{\mathbf{X}}_{t+1:t+Q} \in \mathbb{R}^{Q \times N}$  is the prediction. We decompose the task into node-level local queries. Each node-time pair  $(v, t)$  defines a query with history  $\mathbf{x} = \mathbf{X}_{t-P+1:t,v} \in \mathbb{R}^P$  and future  $\mathbf{y} = \mathbf{X}_{t+1:t+Q,v} \in \mathbb{R}^Q$ , together with the local spatial context of  $v$  induced by  $\mathbf{A}$ . Cross-city few-shot forecasting involves a data-rich source city and a data-scarce target city, denoted by superscripts  $sr$  and  $tg$ . The source city provides a long history  $\mathbf{X}^{sr} \in \mathbb{R}^{T^{sr} \times N^{sr}}$ . The target city offers only a brief span  $\mathbf{X}^{tg} \in \mathbb{R}^{T^{tg} \times N^{tg}}$  with  $T^{tg} \ll T^{sr}$ . Given  $(\mathbf{X}^{sr}, \mathbf{A}^{sr})$  and  $(\mathbf{X}^{tg}, \mathbf{A}^{tg})$ , the goal is to predict the future  $\hat{\mathbf{y}}_q$  of every target-city query  $q = (v, t)$  from history  $\mathbf{x}_q$ , minimizing the error against the ground truth  $\mathbf{y}_q$ .

## 3 Method

### 3.1 Framework Overview

R<sup>2</sup>-Transfer is a multi-agent framework for cross-city few-shot prediction, achieves query-conditioned local knowledge transfer through four specialized agents. As shown in Fig. 3, R<sup>2</sup>-Transfer consists of a Knowledge Base Agent, a Retrieval Agent, a Prediction Agent, and a Reflection Agent.

The Knowledge Base Agent converts the raw spatio-temporal data of the source and target cities into aligned local entries, and

maintains a knowledge base. The Retrieval Agent selects the most relevant entry from the knowledge base for each target query, and passes the entry to the Prediction Agent. The Prediction Agent performs prediction over the target query and the retrieved entry passed from Retrieval Agent, and outputs the prediction value together with the reasoning content. The Reflection Agent attributes a prediction failure to the responsible agent based on the reasoning content, and revises the memory or prompt of the responsible agent.

### 3.2 Knowledge Base Agent

The Knowledge Base Agent converts raw spatio-temporal data of the source and target cities into local entries, and organizes the entries into a high-quality, non-redundant knowledge base. As shown in Fig. 4, the agent performs two main responsibilities. The knowledge base maintenance constructs local entries and decides which entries are kept in knowledge base. And the cross-city spatial alignment maps the spatial information of source-city entries into the target-city space before admission. Together, the two responsibilities convert raw data of different cities into aligned local entries and keep the knowledge base high-quality and non-redundant, which resolves the first challenge on extracting and maintaining transferable knowledge.

**3.2.1 Knowledge Base Maintenance.** The Knowledge Base Agent constructs a local entry and maintains a bounded knowledge base  $\mathcal{B}$  by admitting novel, high-quality entries and discarding redundant entries. The Knowledge Base Agent first segments the spatio-temporal data of each city into segments. Each segment contains the corresponding spatial and temporal knowledge. From each segment, the Knowledge Base Agent builds an entry:

$$b_i = \{\mathbf{x}_i, \mathbf{y}_i, \mathbf{z}_i^{tp}, \mathbf{z}_i^{sp}, \mathbf{c}_i^{tp}, \mathbf{c}_i^{sp}\}, \quad (2)$$

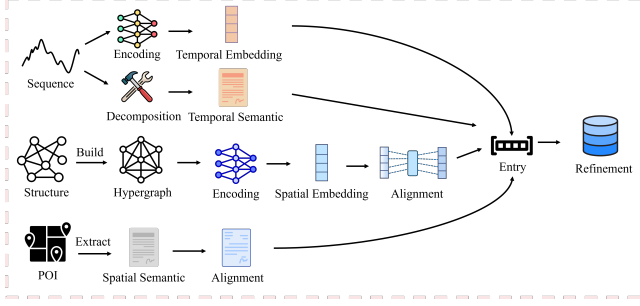


Figure 4: The workflow of the Knowledge Base Agent.

where  $x_i$  and  $y_i$  are the history and the future time series,  $z_i^{tp}$  and  $z_i^{sp}$  are the temporal and spatial embeddings, and  $c_i^{tp}$  and  $c_i^{sp}$  are the temporal and spatial semantics. Specifically, the temporal embedding is produced by a temporal convolutional network (TCN) encoder pre-trained with self-supervised sequence reconstruction. The spatial embedding is produced by a Graph WaveNet encoder pre-trained with graph reconstruction. The spatial encoder operates on a corridor-level hypergraph. This hypergraph construction method classifies nodes with a degree of 1 as endpoint nodes, nodes with a degree of 3 or higher as junction nodes, and nodes with a degree of 2 as intra-edge nodes. The temporal semantics are extracted by applying STL decomposition to  $x_i$  and summarizing the seasonal, trend, and residual components. The spatial semantics summarize nearby Points of Interest with a textual template, and are left empty when POI data are unavailable.

The Knowledge Base Agent then decides whether each candidate entry enters  $\mathcal{B}$ . While  $\mathcal{B}$  is below the capacity, every candidate is admitted. Once  $\mathcal{B}$  is full, the Knowledge Base Agent weighs two factors. The first factor is entry quality: an entry is informative only when its STL decomposition exhibits significant seasonal and trend components rather than dominant noise. The second factor is redundancy: a high temporal-embedding similarity between the candidate and existing entries indicates a duplicated temporal pattern. A candidate replaces the most similar existing entry only when the candidate has higher quality and the KB memory  $\mathcal{M}_{KB}$  records that queries of the corresponding pattern lack defensible analogies; otherwise the candidate is discarded. The record in  $\mathcal{M}_{KB}$  is written by the Reflection Agent, which keeps the admission policy aligned with what target queries actually demand.

**3.2.2 Cross-City Spatial Alignment.** The Knowledge Base Agent aligns the spatial embedding and spatial semantics of each source-city entry with the space of target city before admission. The spatial embeddings of the two cities are produced by independently pre-trained encoders, so their distributions are mismatched. The Knowledge Base Agent applies a residual mapping to each source-city spatial embedding:

$$z_i^{sp} = \tilde{z}_i^{sp} + \text{MLP}(\tilde{z}_i^{sp}), \quad (3)$$

where  $\tilde{z}_i^{sp}$  is the unaligned spatial embedding of the source city. Since no node-level correspondence exists across cities, the mapping is trained with a distribution-level Maximum Mean Discrepancy (MMD) loss between source and target embeddings:

$$\mathcal{L}_{\text{MMD}}(\theta) = \text{MMD}^2\left(\{z_i^{sp, sr}\}_{i=1}^{n_{sr}}, \{z_j^{sp, tg}\}_{j=1}^{n_{tg}}\right), \quad (4)$$

where  $z_i^{sp, sr} = g_\theta(\tilde{z}_i^{sp, sr})$  is the mapped source-city embedding, while  $z_j^{sp, tg}$  is the target-city embedding that defines the reference space. The residual form preserves the structure learned by the pre-trained encoder. For spatial semantics, the Knowledge Base Agent uses an LLM to rewrite each source city descriptor, such as a city-specific POI type, into its closest counterpart in the target city vocabulary. After alignment, source-city and the target-city entries become comparable, so local knowledge can be transferred across cities despite mismatched spatio-temporal patterns.

### 3.3 Retrieval Agent

The Retrieval Agent selects one entry per query through a coarse-to-fine, multi-perspective search. The searching process first retrieves a candidate entry set by temporal similarity from knowledge base. Then the Retrieval Agent reranks the entries from temporal and spatial perspectives, and finally selects one entry under a weight maintained in the retrieval memory  $\mathcal{M}_{RT}$ . The selected entry serves as the augmented entry  $b_{\text{aug}}$  and is passed to the Prediction Agent.

In the coarse searching stage, the Retrieval Agent takes the temporal embedding of the query entry  $b_q$  and performs ANN search over the knowledge base  $\mathcal{B}$ . The search returns the top- $k$  most similar entries as the candidate set. Specifically, the candidate set is defined as:

$$C_q = \text{TopK}_{b_i \in \mathcal{B}}\left(\cos(z_q^{tp}, z_i^{tp}), k\right), \quad (5)$$

where  $k$  controls the candidate-set size. Temporal similarity offers an efficient first filter because the history time series is the basis of prediction.

In the fine searching stage, the Retrieval Agent reranks the candidate entry set independently from three perspectives: temporal semantics, spatial semantics, and spatial embedding. For the two semantic perspectives, entries are ranked by the number of matched semantic fields. For the spatial embedding perspective, entries are ranked by cosine similarity. Each perspective keeps its top-ranked entry. Independent reranking preserves the best evidence of each perspective, since temporal relevance and spatial relevance may favor different entries.

Finally, the Retrieval Agent chooses  $b_{\text{aug}}$  among the per-perspective entries according to a weight stored in  $\mathcal{M}_{RT}$ . The profile assigns each perspective a weight level, and the entry nominated by the highest weighted perspective is selected. The Reflection Agent revises the level of weights by reflecting based on reasoning content. The coarse-to-fine, multi-perspective searching process resolves the second challenge: temporal relevance and spatial relevance are examined separately, so only the genuinely useful knowledge is selected for each query.

### 3.4 Prediction Agent

The Prediction Agent performs analogical prediction over the target query and the retrieved entry, and outputs the prediction value together with a reasoning content. Specifically, the Prediction Agent takes three inputs: the query entry  $b_q$ , the augmented entry  $b_{\text{aug}}$  passed from the Retrieval Agent, and the prediction memory  $\mathcal{M}_P$ . The history time series of  $b_q$  provides the basis of prediction. The reference future of  $b_{\text{aug}}$  serves as the analogical evidence.  $\mathcal{M}_P$  records error patterns from past predictions and prevents repeated misuse

of unreliable references. Based on the three inputs, the Prediction Agent produces the prediction  $\hat{y}_q$  for the target query. Along with the prediction, the Prediction Agent generates a reasoning content  $\psi_q$ . The reasoning content states which fields of  $b_{\text{aug}}$  support each step of the prediction. The Reflection Agent relies on  $\psi_q$  to attribute a prediction failure to irrelevant retrieval or mistaken analogical reasoning.

### 3.5 Reflection Agent

The Reflection Agent diagnoses and attributes prediction failure to the responsible agent, and revises the memory or prompt of that agent through bounded and reversible updates. A failure may stem from missing knowledge in the knowledge base, from mismatched retrieval, or from biased prediction. The three causes require revisions on different agents. Therefore, the Reflection Agent works in two stages: failure diagnosis and reflection with rollback.

**3.5.1 Failure Diagnosis.** The Reflection Agent monitors prediction errors and classifies each failure into one of three root causes. Reflection is triggered only when the current prediction error exceeds an exponential moving average (EMA) of recent errors:

$$\mathbb{I}_{\text{reflect}}(q) = \mathbb{I}[\epsilon_q > (1 + \gamma) \bar{\epsilon}], \quad (6)$$

where  $\gamma$  is a relative-margin parameter, and  $\bar{\epsilon}$  is the EMA of recent errors. The relative trigger avoids reflecting on errors that are large in absolute scale but consistent with the recent error level. Once triggered, the Reflection Agent reads the reasoning content  $\psi_q$  to attribute the failure. The failure is classified as knowledge base insufficiency when the cited fields are judged uninformative in  $\psi_q$  and similar judgments have recurred recently. The failure is classified as retrieval mismatch when  $\psi_q$  reveals that a wrong retrieval perspective drove the selection of the augmented entry. The failure is classified as prediction bias when the augmented entry is coherently used but the prediction still drifts beyond what the analogy warrants. The diagnosis yields a single update target for the reflection stage.

**3.5.2 Reflection and Rollback.** The Reflection Agent revises the responsible agent according to the diagnosis, and reverts the revision if subsequent predictions do not improve. For knowledge base insufficiency, the Reflection Agent writes into  $\mathcal{M}_{KB}$  the query pattern that lacks defensible analogies. The Knowledge Base Agent then relaxes the redundancy threshold for entries matching the missing pattern in later admission decisions. For retrieval mismatch, the Reflection Agent revises the perspective weights in  $\mathcal{M}_{RT}$ . The perspective that nominated the augmented entry is downgraded, and the perspective identified as more defensible in  $\psi_q$  is upgraded. For prediction bias, the Reflection Agent writes the uncovered error pattern into  $\mathcal{M}_p$ . If the bias reflects a general flaw in using retrieved entry, the Reflection Agent instead replaces a short instruction in the prompt of the prediction agent. Each proposed revision is validated against bounded-update constraints: format validity, weight-adjustment bounds, and prompt safety rules. Revisions failing validation are discarded. Each applied modification then enters an observation window over the next several queries. If the average error within the window does not fall below the error that triggered the reflection, the modification is reverted.

## 4 Experiment

In this section, we evaluate the effectiveness of the proposed R<sup>2</sup>-Transfer framework for cross-city few-shot prediction. We begin by detailing the experimental setup, including dataset descriptions, metrics, and baseline comparisons. To validate the advantages of R<sup>2</sup>-Transfer, we conduct a comprehensive performance comparison against state-of-the-art methods. We further design ablation experiments, hyperparameter analyses, knowledge base analysis, case study, explainability analysis, and reflection process analysis to examine the contribution of each component in R<sup>2</sup>-Transfer.

### 4.1 Experimental Setup

**4.1.1 Datasets.** We evaluate R<sup>2</sup>-Transfer on three source-to-target city pairs. For traffic prediction without POIs, the pairs are METR-LA to PEMS-BAY and Didi-Shenzhen to Didi-Chengdu. For bike-demand prediction with POIs, we use NY Bike as the source and DC Bike as the target. To simulate few-shot transfer, each target dataset is restricted to its first three training days. Both the historical input and prediction horizon contain 12 time steps. Detailed dataset statistics and preprocessing are provided in Appendix A.1.

**4.1.2 Baselines and Evaluation Metrics.** We compare R<sup>2</sup>-Transfer with HA, ARIMA [37], DCRNN [16], GWN [38], DSTAGNN [14], FOGS [29], TransGTR [13], TPB [21], GPD [44], and XRAG4TS [10]. We evaluate all methods using Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE). Descriptions of the baselines and metric definitions are provided in Appendix A.2.

### 4.2 Performance of R<sup>2</sup>-Transfer

**4.2.1 Overall Comparison.** We evaluate the performance of R<sup>2</sup>-Transfer against a diverse set of baselines. The experimental results are shown in Table. 1. Across all evaluations, R<sup>2</sup>-Transfer achieves the best performance on 15 out of the 18 reported evaluations across M-P, S-C, and N-D.

Compared with TPB, R<sup>2</sup>-Transfer reduces the 60 mins time step RMSE by 24.13% on M-P. The improvement benefits from transferring knowledge as local entries instead of compressing source-city knowledge into a traffic pattern bank. Compared with GPD, R<sup>2</sup>-Transfer reduces the 60 mins time step RMSE by 23.01% on M-P. Query-conditioned local retrieval selects relevant source-city knowledge for each target query, while generative pre-training transfers source-city knowledge through globally learned model parameters. Besides, R<sup>2</sup>-Transfer surpasses XRAG4TS at the 60 mins time step on M-P, reducing MAE and RMSE by 17.47% and 17.00%, respectively. XRAG4TS retrieves relevant temporal patterns but does not explicitly model cross-city spatial alignment or prediction feedback. R<sup>2</sup>-Transfer further combines cross-city spatial alignment, multi-perspective retrieval, and reflection-driven adaptation, providing more relevant and reliable local knowledge for long-term prediction.

### 4.3 Ablation Study

To quantify the contribution of each core component in R<sup>2</sup>-Transfer, we conduct an ablation study by comparing the full framework against four variants: **w/o RAG Retrieval** (removes the knowledge

**Table 1: Performance comparison on METR-LA  $\rightarrow$  PEMS-BAY (M-P), Didi-Shenzhen  $\rightarrow$  Didi-Chengdu (S-C), and NY Bike  $\rightarrow$  DC Bike (N-D). Each metric cell reports the mean above its standard deviation. The best and second-best means are highlighted in bold and underlined, respectively. Improve reports the relative error reduction over the best competing method.**

| Method                        | M-P         |             |             |             |             |             | S-C         |             |             |             |             |             | N-D         |             |             |             |             |             |
|-------------------------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
|                               | 5 min       |             | 15 min      |             | 60 min      |             | 10 min      |             | 30 min      |             | 120 min     |             | 1 hour      |             | 3 hours     |             | 12 hours    |             |
|                               | MAE         | RMSE        | MAE         | RMSE        | MAE         | RMSE        | MAE         | RMSE        | MAE         | RMSE        | MAE         | RMSE        | MAE         | RMSE        | MAE         | RMSE        | MAE         | RMSE        |
| HA                            | 2.98        | 6.89        | 2.92        | 6.78        | 5.23        | 9.80        | 2.85        | 3.99        | 3.28        | 4.69        | 5.33        | 7.06        | 2.24        | 4.49        | 2.59        | 4.30        | 2.77        | 4.79        |
| ARIMA                         | $\pm 0.66$  | $\pm 1.49$  | $\pm 0.40$  | $\pm 1.56$  | $\pm 0.46$  | $\pm 2.23$  | $\pm 0.32$  | $\pm 0.71$  | $\pm 0.56$  | $\pm 0.78$  | $\pm 0.39$  | $\pm 0.54$  | $\pm 0.36$  | $\pm 0.87$  | $\pm 0.55$  | $\pm 1.13$  | $\pm 0.72$  | $\pm 0.94$  |
|                               | 2.08        | 4.30        | 2.41        | 5.12        | 5.00        | 9.43        | 2.81        | 3.86        | 3.25        | 4.52        | 5.23        | 6.80        | 2.04        | 3.28        | 2.46        | 3.89        | 5.00        | 9.43        |
|                               | $\pm 0.11$  | $\pm 0.38$  | $\pm 0.16$  | $\pm 0.60$  | $\pm 0.26$  | $\pm 1.21$  | $\pm 0.29$  | $\pm 0.68$  | $\pm 0.27$  | $\pm 0.42$  | $\pm 0.69$  | $\pm 0.86$  | $\pm 0.45$  | $\pm 0.84$  | $\pm 0.54$  | $\pm 0.60$  | $\pm 1.04$  | $\pm 0.91$  |
| DCRNN                         | 1.43        | 2.34        | 1.89        | 3.87        | 3.40        | 6.08        | 2.29        | 3.30        | 2.87        | 4.11        | 4.16        | 5.75        | 1.27        | 2.67        | 1.74        | 3.54        | 2.21        | 4.20        |
|                               | $\pm 0.07$  | $\pm 0.08$  | $\pm 0.12$  | $\pm 0.10$  | $\pm 0.13$  | $\pm 0.17$  | $\pm 0.05$  | $\pm 0.08$  | $\pm 0.04$  | $\pm 0.08$  | $\pm 0.15$  | $\pm 0.19$  | $\pm 0.08$  | $\pm 0.10$  | $\pm 0.05$  | $\pm 0.60$  | $\pm 0.09$  | $\pm 0.06$  |
| GWN                           | 1.44        | 2.39        | 1.94        | 3.88        | 3.20        | 6.13        | 2.44        | 3.32        | 2.95        | 4.21        | 4.22        | 5.80        | 1.23        | 2.62        | 1.72        | 3.54        | 2.12        | 4.47        |
|                               | $\pm 0.08$  | $\pm 0.12$  | $\pm 0.10$  | $\pm 0.11$  | $\pm 0.06$  | $\pm 0.09$  | $\pm 0.06$  | $\pm 0.10$  | $\pm 0.12$  | $\pm 0.16$  | $\pm 0.12$  | $\pm 0.14$  | $\pm 0.08$  | $\pm 0.11$  | $\pm 0.10$  | $\pm 0.16$  | $\pm 0.12$  | $\pm 0.09$  |
| DSTAGNN                       | 1.33        | 2.21        | 1.85        | 3.35        | 3.35        | 6.25        | 2.45        | 3.46        | 2.94        | 4.11        | 4.37        | 5.98        | 1.49        | 2.75        | 1.81        | 3.58        | 2.30        | 4.59        |
|                               | $\pm 0.11$  | $\pm 0.17$  | $\pm 0.21$  | $\pm 0.16$  | $\pm 0.24$  | $\pm 0.20$  | $\pm 0.13$  | $\pm 0.16$  | $\pm 0.17$  | $\pm 0.20$  | $\pm 0.23$  | $\pm 0.27$  | $\pm 0.10$  | $\pm 0.12$  | $\pm 0.12$  | $\pm 0.11$  | $\pm 0.16$  | $\pm 0.12$  |
| FOGS                          | 1.24        | 2.22        | 1.82        | 3.30        | 3.22        | 6.19        | 2.30        | 3.29        | 2.86        | 4.13        | 4.17        | 5.81        | 1.35        | 2.71        | 1.79        | 3.23        | 2.06        | 4.12        |
|                               | $\pm 0.09$  | $\pm 0.07$  | $\pm 0.08$  | $\pm 0.13$  | $\pm 0.12$  | $\pm 0.15$  | $\pm 0.09$  | $\pm 0.11$  | $\pm 0.08$  | $\pm 0.11$  | $\pm 0.12$  | $\pm 0.13$  | $\pm 0.07$  | $\pm 0.06$  | $\pm 0.08$  | $\pm 0.08$  | $\pm 0.10$  | $\pm 0.07$  |
| TransGTR                      | 1.12        | 2.14        | 1.78        | 3.38        | 3.26        | 5.64        | 2.26        | 3.21        | 2.72        | 4.17        | 3.73        | 5.65        | 1.18        | 2.47        | 1.65        | <b>2.93</b> | 1.98        | 3.64        |
|                               | $\pm 0.03$  | $\pm 0.08$  | $\pm 0.10$  | $\pm 0.14$  | $\pm 0.17$  | $\pm 0.30$  | $\pm 0.10$  | $\pm 0.36$  | $\pm 0.15$  | $\pm 0.21$  | $\pm 0.23$  | $\pm 0.31$  | $\pm 0.17$  | $\pm 0.09$  | $\pm 0.10$  | $\pm 0.13$  | $\pm 0.11$  | $\pm 0.08$  |
| TPB                           | 1.09        | 1.91        | 1.60        | 3.16        | 3.28        | 5.47        | 2.13        | 3.08        | 2.67        | 3.84        | 3.49        | 5.04        | 1.23        | 2.45        | 1.60        | 3.11        | 2.00        | 3.82        |
|                               | $\pm 0.06$  | $\pm 0.09$  | $\pm 0.14$  | $\pm 0.15$  | $\pm 0.22$  | $\pm 0.19$  | $\pm 0.09$  | $\pm 0.11$  | $\pm 0.08$  | $\pm 0.10$  | $\pm 0.10$  | $\pm 0.08$  | $\pm 0.07$  | $\pm 0.06$  | $\pm 0.07$  | $\pm 0.09$  | $\pm 0.12$  | $\pm 0.10$  |
| GPD                           | <u>1.07</u> | 1.96        | 1.56        | 3.23        | 3.12        | 5.39        | 2.18        | 3.12        | 2.54        | 3.76        | 3.51        | <u>4.77</u> | <u>1.13</u> | 2.40        | <u>1.53</u> | 3.00        | 1.98        | 4.06        |
|                               | $\pm 0.05$  | $\pm 0.04$  | $\pm 0.08$  | $\pm 0.10$  | $\pm 0.12$  | $\pm 0.23$  | $\pm 0.05$  | $\pm 0.03$  | $\pm 0.04$  | $\pm 0.07$  | $\pm 0.08$  | $\pm 0.15$  | $\pm 0.04$  | $\pm 0.05$  | $\pm 0.05$  | $\pm 0.09$  | $\pm 0.14$  | $\pm 0.11$  |
| XRAG4TS                       | <b>1.03</b> | <u>1.88</u> | <u>1.49</u> | <u>3.02</u> | <u>2.92</u> | <u>5.00</u> | <b>1.90</b> | <u>3.01</u> | <u>2.43</u> | <u>3.67</u> | <u>3.25</u> | 4.90        | 1.14        | <u>2.33</u> | <u>1.53</u> | 3.04        | <u>1.90</u> | <u>3.62</u> |
|                               | $\pm 0.07$  | $\pm 0.06$  | $\pm 0.09$  | $\pm 0.15$  | $\pm 0.14$  | $\pm 0.21$  | $\pm 0.05$  | $\pm 0.05$  | $\pm 0.08$  | $\pm 0.09$  | $\pm 0.15$  | $\pm 0.23$  | $\pm 0.07$  | $\pm 0.14$  | $\pm 0.10$  | $\pm 0.14$  | $\pm 0.15$  | $\pm 0.18$  |
| <b>R<sup>2</sup>-Transfer</b> | <u>1.07</u> | <b>1.82</b> | <b>1.34</b> | <b>2.53</b> | <b>2.41</b> | <b>4.15</b> | <u>2.06</u> | <b>2.96</b> | <b>2.39</b> | <b>3.42</b> | <b>2.94</b> | <b>4.28</b> | <b>1.10</b> | <b>2.26</b> | <b>1.52</b> | <u>2.97</u> | <b>1.81</b> | <b>3.53</b> |
|                               | $\pm 0.06$  | $\pm 0.10$  | $\pm 0.10$  | $\pm 0.13$  | $\pm 0.18$  | $\pm 0.27$  | $\pm 0.11$  | $\pm 0.09$  | $\pm 0.15$  | $\pm 0.11$  | $\pm 0.21$  | $\pm 0.26$  | $\pm 0.05$  | $\pm 0.07$  | $\pm 0.06$  | $\pm 0.11$  | $\pm 0.14$  | $\pm 0.18$  |
| <b>Improve</b>                | –           | +3.19%      | +10.07%     | +16.23%     | +17.47%     | +17.00%     | –           | +1.66%      | +1.65%      | +6.81%      | +9.54%      | +10.27%     | +2.65%      | +3.00%      | +0.65%      | –           | +4.74%      | +2.49%      |

base), **w/o Semantic Reranking**(removes the reranking perspective of spatio-temporal semantic), **w/o Spatial Alignment**(removes the MMD-based cross-city projection), **w/o Reflection**(removes the reflection mechanism). As shown in Table. 2, removing any major component degrades performance, confirming the effectiveness of each design. On M-P, removing spatial alignment causes the largest average deterioration, demonstrating the importance of making source and target city spatial representations comparable before local knowledge transfer. Removing local retrieval and reflection also leads to substantial long-term prediction degradation, indicating that retrieved entry provides essential knowledge and that feedback-driven updates help correct accumulated retrieval and prediction errors.

#### 4.4 Hyperparameter Analysis

We further conduct hyperparameter analysis for the capacity of knowledge base and the number of coarse retrieved entries  $N_{\text{ret}}$  on M-P. As shown in Fig. 6, the coarse candidate-set size increasing  $N_{\text{ret}}$  from 1 to 3 improves both MAE and RMSE at every prediction horizon. A candidate set containing only one entry provides limited coverage for multi-perspective reranking. An overly large candidate set lead to encountering entries with low matching accuracy or redundancy. Although  $N_{\text{ret}} = 5$  and  $N_{\text{ret}} = 7$  achieve isolated improvements,  $N_{\text{ret}} = 3$  provides stable accuracy with lower retrieval cost.

For the knowledge base capacity, increasing  $|\mathcal{B}|_{\text{max}}$  from 0 to 50K entries improves performance. Performance saturates beyond 50K entries. A capacity of 100K entries provides marginal gains for several short-term metrics but increases the errors at the 30

**Table 2: Ablation results on M-P and N-D. Steps 1/3/6/12 correspond to 5/15/30/60 mins on M-P and 1/3/6/12 hours on N-D.**

| Variant                       | M-P         |             |             |             |             |             |             |             |
|-------------------------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
|                               | Step 1      |             | Step 3      |             | Step 6      |             | Step 12     |             |
|                               | MAE         | RMSE        | MAE         | RMSE        | MAE         | RMSE        | MAE         | RMSE        |
| w/o RAG                       | 1.07        | 1.88        | 1.56        | 3.30        | 1.88        | 4.05        | 3.17        | 5.24        |
| w/o Sem. Rerank.              | 1.08        | 1.84        | 1.36        | 2.85        | 1.65        | 3.22        | 2.47        | 4.44        |
| w/o Reflection                | 1.07        | 1.88        | 1.41        | 2.93        | 1.99        | 4.27        | 3.04        | 5.25        |
| w/o Spatial Align.            | 1.10        | 1.95        | 1.54        | 3.16        | 2.07        | 4.30        | 3.09        | 5.13        |
| <b>R<sup>2</sup>-Transfer</b> | <b>1.07</b> | <b>1.82</b> | <b>1.34</b> | <b>2.53</b> | <b>1.54</b> | <b>3.36</b> | <b>2.41</b> | <b>4.15</b> |
| Variant                       | N-D         |             |             |             |             |             |             |             |
|                               | Step 1      |             | Step 3      |             | Step 6      |             | Step 12     |             |
|                               | MAE         | RMSE        | MAE         | RMSE        | MAE         | RMSE        | MAE         | RMSE        |
| w/o RAG                       | 1.15        | 2.37        | 1.58        | 3.14        | 1.62        | 3.45        | 1.99        | 3.85        |
| w/o Sem. Rerank.              | 1.13        | 2.32        | 1.60        | 3.15        | 1.65        | 3.53        | 1.93        | 3.94        |
| w/o Reflection                | 1.10        | 2.35        | 1.61        | 3.17        | 1.68        | 3.48        | 2.00        | 3.83        |
| w/o Spatial Align.            | 1.18        | 2.29        | 1.58        | 3.08        | 1.67        | 3.50        | 1.97        | 3.88        |
| <b>R<sup>2</sup>-Transfer</b> | <b>1.10</b> | <b>2.26</b> | <b>1.52</b> | <b>2.97</b> | <b>1.53</b> | <b>3.22</b> | <b>1.81</b> | <b>3.53</b> |

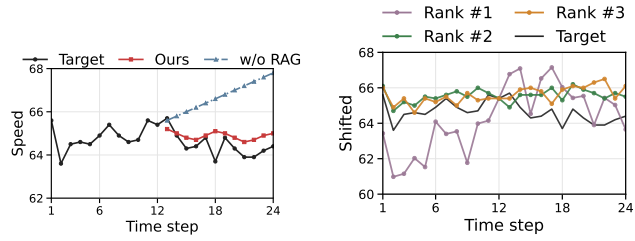
and 60 minute. Removing the capacity constraint further degrades performance. An undersized knowledge base cannot cover sufficient local patterns, whereas an unbounded knowledge base retains more redundant or low-quality entries and weakens retrieval reliability.

#### 4.5 Knowledge Base Analysis

Fig. 7 shows the four largest knowledge-base groups on M-P; the remaining four are shown in Appendix B.1. Across the eight groups, representativeness exceeds 0.82, quality scores remain around 0.80, and value scores range from 0.48 to 0.56. The alignment confidence

**Table 3: Reflection process statistics. Trigger rate is the fraction of prediction queries that activate the Reflection Agent. Applied rate is computed over triggered reflections; proposed revisions that fail the bounded-update validation are discarded before application. Keep and rollback rates are computed over applied revisions. Target columns report the distribution of applied revision targets.**

| Setting | Reflection Behavior |         |        |          | Revision Target Distribution |           |            |              |
|---------|---------------------|---------|--------|----------|------------------------------|-----------|------------|--------------|
|         | Trigger             | Applied | Keep   | Rollback | KB Mem.                      | Ret. Mem. | Pred. Mem. | Pred. Prompt |
| M-P     | 14.91%              | 97.04%  | 81.04% | 18.96%   | 2.1%                         | 15.1%     | 46.9%      | 36.0%        |
| N-D     | 10.51%              | 90.74%  | 85.81% | 14.19%   | 1.2%                         | 18.8%     | 44.5%      | 35.5%        |



**(a) Comparison of the target future with predictions from R<sup>2</sup>-Transfer and w/o RAG. (b) Top retrieved entries shifted to match the last target-history value.**

**Figure 5: Case study on M-P.**

of source-city groups (0.72–0.79) also overlaps that of target-city groups (0.74–0.81). These results indicate that the knowledge base preserves diverse local patterns and makes aligned source entries as retrievable as target entries.

#### 4.6 Case Study

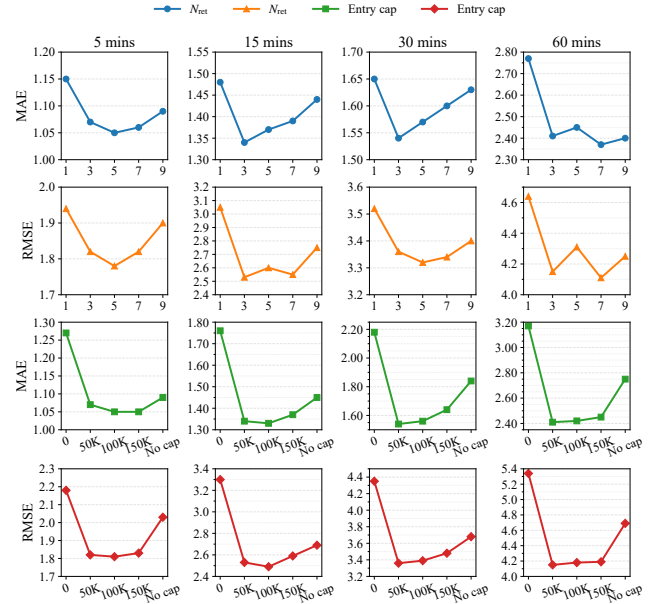
We study a query from node 259 in PEMS-BAY. R<sup>2</sup>-Transfer reduces its 12-step MAE from 2.48 to 1.38 relative to w/o RAG Retrieval. Although the retrieved entries in Fig. 5b have similar histories, their futures diverge, so the Retrieval Agent uses temporal semantics and aligned spatial information to select the most relevant analogy. As shown in Fig. 5a, the selected entry helps R<sup>2</sup>-Transfer avoid the severe downward bias of the ablation variant. The case demonstrates query-conditioned local transfer and an inspectable retrieval-to-prediction path. Further analysis is provided in Appendix B.2.

#### 4.7 Explainability

R<sup>2</sup>-Transfer records an explicit reasoning trace for each prediction, making the use of retrieved knowledge inspectable and providing evidence for failure attribution. In a representative M-P trace, the Prediction Agent identifies a poor temporal match caused by a five-hour offset but a reliable spatial match, discounts a weak short-term rise in favor of a stable retrieved trajectory, and determines that no stored error pattern applies. The trace therefore explains both the final stable forecast and why no error compensation is used. The full reasoning output and detailed interpretation are provided in Appendix B.3.

#### 4.8 Reflection Process Analysis

Table. 3 shows that reflection is selective: only 14.91% of M-P and 10.51% of N-D queries trigger it. More than 90% of triggered revisions are applied, and over 81% of applied revisions are retained.



**Figure 6: Hyperparameter sensitivity across time steps on M-P. The first two rows vary the number of coarse retrieved entries  $N_{ret}$ , and the last two rows vary the knowledge base capacity.**

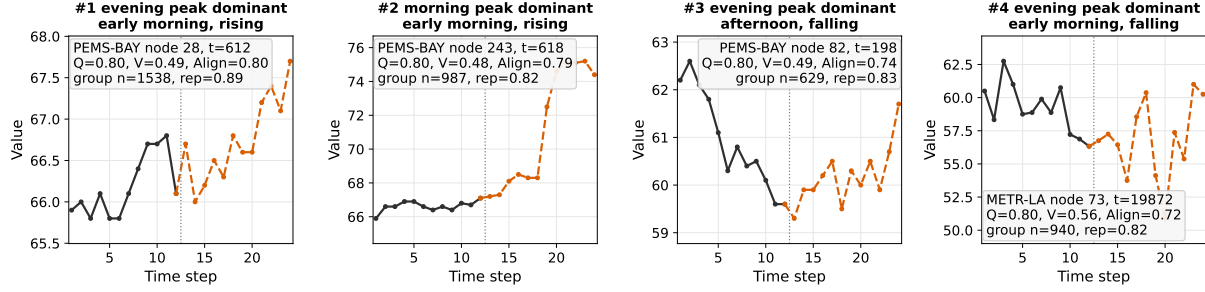
Table. 5 further shows that retained revisions recover about 63% of the triggering error and return window MAE to within 0.6 of the pre-update EMA baseline. Rolled-back revisions instead exceed that baseline by more than 5.0 and never outperform it. These results support both the effectiveness and stability of bounded reflection. Detailed analysis is provided in Appendix B.4.

#### 4.9 Efficiency Comparison

Table. 4 confirms that R<sup>2</sup>-Transfer is portable across four reasoning LLMs. GPT o1 achieves the best result in seven of eight evaluations but costs \$92.75 per 1,000 queries. Claude Sonnet 4 remains close in accuracy while reducing cost by 77.2%, DeepSeek-R1 lowers cost to \$3.39 with moderate degradation, and Gemini is the fastest and least expensive but least accurate. The framework therefore supports different accuracy, latency, and cost priorities. Detailed comparisons are provided in Appendix B.5.

**Table 4: Prediction performance and service-level efficiency of R<sup>2</sup>-Transfer with different reasoning LLM backbones on M-P.**

| Backbone                      | Prediction Performance |             |             |             |             |             |             |             | Service-Level Efficiency     |                               |                            |                         |
|-------------------------------|------------------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|------------------------------|-------------------------------|----------------------------|-------------------------|
|                               | 5 mins                 |             | 15 mins     |             | 30 mins     |             | 60 mins     |             | Input Price<br>(\$/M tokens) | Output Price<br>(\$/M tokens) | Latency/Query<br>(seconds) | Cost/1K Queries<br>(\$) |
|                               | MAE                    | RMSE        | MAE         | RMSE        | MAE         | RMSE        | MAE         | RMSE        |                              |                               |                            |                         |
| GPT o1                        | <b>1.07</b>            | <b>1.82</b> | <b>1.34</b> | <b>2.53</b> | <b>1.54</b> | <b>3.36</b> | <b>2.41</b> | <b>4.15</b> | 15.00                        | 60.00                         | 30.2                       | 92.75                   |
| DeepSeek-R1                   | <u>1.09</u>            | 1.88        | 1.40        | 2.64        | 1.66        | 3.51        | 2.57        | 4.36        | 0.55                         | 2.19                          | 48.8                       | 3.39                    |
| Claude Sonnet 4               | <b>1.07</b>            | <b>1.80</b> | <u>1.37</u> | <u>2.62</u> | <u>1.59</u> | <u>3.48</u> | <u>2.43</u> | <u>4.20</u> | 3.00                         | 15.00                         | 43.7                       | 21.15                   |
| Gemini 2.0 Flash Experimental | 1.15                   | 2.03        | 1.52        | 3.11        | 1.91        | 4.10        | 3.23        | 5.37        | 0.35                         | 0.60                          | <b>28.1</b>                | <b>1.47</b>             |

**Figure 7: Representative entries of the four largest knowledge-base groups under the M-P setting.****Table 5: Observation-window error change after applied revisions. Drop vs. Trigger measures how much of the triggering error is recovered within the window.  $\Delta$  vs. EMA compares the window MAE with the reconstructed pre-update EMA baseline (positive means above the normal error level); Win Rate is the fraction of windows whose MAE falls below this baseline. Rollback rows are measured before reversion.**

| Setting | Verdict  | Drop vs. Trigger | $\Delta$ vs. EMA | Win Rate |
|---------|----------|------------------|------------------|----------|
| M-P     | Keep     | 62.92%           | +0.58            | 38.95%   |
| M-P     | Rollback | -106.94%         | +5.07            | 0.00%    |
| N-D     | Keep     | 62.95%           | +0.60            | 46.61%   |
| N-D     | Rollback | -59.86%          | +5.54            | 0.00%    |

## 5 Related Work

### 5.1 Cross-City Spatio-Temporal Forecasting

Deep spatio-temporal forecasting models capture temporal and spatial dependencies through recurrent networks, graph convolutions, adaptive graphs, and dynamic graph learning [14, 16, 29, 30, 38, 42]. However, their strong performance usually relies on sufficient target-city observations. Cross-city forecasting alleviates this requirement by transferring knowledge from data-rich cities. Existing methods mainly follow four directions. First, region alignment, meta-learning, and domain adaptation transfer representations or reduce distribution gaps across cities, as in RegionTrans, MetaST, STAN, DASTNet, and CrossTReS [3, 12, 28, 33, 35, 40]. Second, few-shot methods reconstruct target graphs, learn transferable meta-knowledge, or generate prediction networks [23, 44]. Third, TransGTR transfers graph structures, while TPB and MTPB summarize recurrent traffic patterns into single- or multi-scale pattern banks [13, 21, 22]. Fourth, pFedCTP, STBaT, UniST, FlashST, and OpenCity improve generalization through federated learning, balanced or large-scale pre-training, and prompt tuning [6, 17, 19, 43, 46]. Nevertheless, most methods transfer source-city knowledge

through global, dataset-level, or region-level carriers, such as model parameters, latent representations, graph structures, and pattern prototypes. Although region weighting and pattern banks reduce irrelevant transfer, the knowledge required by a specific prediction is often local and query-dependent. A source entry is useful only when both its temporal and spatial patterns match the target query; otherwise, it may introduce noise and degrade performance. Related LLM-based, retrieval-augmented, and agent-oriented forecasting studies are reviewed in Appendix C.1.

### 5.2 LLM for Spatio-Temporal Forecasting

LLM-based forecasting represents numerical sequences as text, adapts pretrained language models, or incorporates spatial and urban knowledge [1, 2, 4, 7–9, 11, 18, 26, 27, 32, 47, 48]. Retrieval-augmented methods introduce relevant historical patterns [5, 10, 15, 20, 25, 34, 39, 45], while agent-based methods enable reasoning and reflection [24, 31, 36, 41]. However, existing pipelines remain monolithic, making prediction failures difficult to attribute to knowledge construction, retrieval, or utilization.

## 6 Conclusion and Discussion

We presented R<sup>2</sup>-Transfer, a reflective retrieval-augmented multi-agent framework for cross-city few-shot prediction. It preserves source-city experience as aligned local entries, retrieves query-relevant evidence, predicts by analogy, and applies bounded reflection without retraining. Experiments on three city pairs show consistent improvements, especially at long horizons. Two limitations remain: retrieval depends on the coverage and quality of source entries and metadata. Noisy metadata, missing POIs, or large differences in sensing infrastructure and mobility patterns may weaken the retrieved analogies. LLM agents also add inference cost and variability across backbones or decoding settings. Future work will improve robustness to source-target mismatch and reduce inference cost.

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## A Experimental Details

### A.1 Datasets

We use two traffic dataset pairs without POIs and one bike-demand dataset pair with POIs. METR-LA is the source dataset and PEMS-BAY is the target dataset in the first traffic setting; Didi-Shenzhen is the source dataset and Didi-Chengdu is the target dataset in the second. For the bike-demand setting with POIs, NY Bike and DC Bike serve as the source and target datasets, respectively.

METR-LA and PEMS-BAY use 5-minute intervals, Didi-Chengdu and Didi-Shenzhen use 10-minute intervals, and NY Bike and DC Bike use 1-hour intervals. We process each source dataset chronologically: entries constructed from the first 90% of the data are directly admitted to the knowledge base by the Knowledge Base Agent, while the remaining 10% serves as warm-up data and undergoes the full testing process. To simulate the few-shot setting, we restrict each target dataset to the first three days of its training set. Both the input and output windows contain 12 time steps.

### A.2 Baselines and Evaluation Metrics

We compare R<sup>2</sup>-Transfer with the following baselines:

- **HA** is a classical statistical baseline that predicts future traffic conditions using the historical average of observations at the same or similar time periods.
- **ARIMA** [37] models temporal dependencies using autoregression, differencing, and moving-average components.
- **DCRNN** [16] models traffic dynamics as a diffusion process on a directed graph and integrates diffusion convolution with recurrent neural networks.
- **GWN** [38] combines graph convolution with dilated causal convolutions for spatio-temporal forecasting.
- **DSTAGNN** [14] is a dynamic spatio-temporal aware graph neural network for traffic flow forecasting.
- **FOGS** [29] introduces first-order gradient supervision and a learning-based graph construction module for traffic flow forecasting.
- **TransGTR** [13] is a transferable graph structure learning framework for cross-city traffic forecasting.
- **TPB** [21] performs cross-city few-shot traffic forecasting by constructing a traffic pattern bank from data-rich source cities.

**Table 6: Statistics of the traffic datasets used in the experiments.**

| Datasets  | METR-LA | PEMS-BAY | Didi-Chengdu | Didi-Shenzhen |
|-----------|---------|----------|--------------|---------------|
| # Nodes   | 207     | 325      | 524          | 627           |
| # Edges   | 1,722   | 2,694    | 1,120        | 4,845         |
| Interval  | 5 min   | 5 min    | 10 min       | 10 min        |
| Time span | 34,272  | 52,116   | 17,280       | 17,280        |
| Mean      | 58.274  | 61.776   | 29.023       | 31.001        |
| Std       | 13.128  | 9.285    | 9.662        | 10.969        |

**Table 7: Statistics of the bike-demand datasets. Longitude and latitude indicate the spatial ranges of the selected datasets.**

| City Abbr.      | Longitude (W) $W_c$ | Latitude (N) $H_c$ | Time            | # Bikes |
|-----------------|---------------------|--------------------|-----------------|---------|
| New York (NY)   | [74.059, 73.863]    | [40.645, 40.848]   | 1/1–31/12, 2016 | 13.8M   |
| Washington (DC) | [77.127, 76.926]    | [38.798, 38.969]   | 1/1–31/12, 2016 | 2.7M    |

• **GPD [44]** is a generative pre-training framework for spatio-temporal few-shot learning under urban data scarcity.

• **XRAG4TS [10]** is a retrieval-augmented LLM framework that selectively transfers relevant temporal patterns from data-rich source cities for cross-city forecasting.

We use MAE and RMSE to measure the discrepancy between the multi-step predictions and ground truth. They are defined as

$$\text{MAE}(y, \hat{y}) = \frac{1}{N} \sum_{i=1}^N |\hat{y}_i - y_i|, \quad (7)$$

and

$$\text{RMSE}(y, \hat{y}) = \sqrt{\frac{1}{N} \sum_{i=1}^N (\hat{y}_i - y_i)^2}, \quad (8)$$

where  $y_i$  is the ground-truth value,  $\hat{y}_i$  is the corresponding prediction, and  $N$  is the total number of predicted samples.

## B Additional Experimental Analysis

### B.1 Knowledge Base Analysis

Fig. 7 and Fig. 8 together show the eight most populous entry groups under the M-P setting, grouped by temporal-embedding similarity. Each panel plots the representative entry of one group and reports five statistics:  $Q$  is the entry quality score derived from STL decomposition,  $V$  is the composite value score,  $\text{Align}$  is the confidence after cross-city spatial alignment,  $n$  is the group size, and  $\text{Rep}$  is the representativeness of the plotted entry.

The visualization provides three further observations. First, the groups retain distinct temporal regimes instead of collapsing source experience into a global summary. Their consistently high quality and representativeness confirm that the admission policy maintains a compact, nonredundant knowledge base. Second, temporal semantics distinguish entries with similar histories but different futures. Groups #1 and #2, for example, have nearly flat early-morning histories but different semantic labels and future trajectories: the morning-peak-dominant group surges as the peak approaches, whereas the evening-peak-dominant group rises only mildly. Third, aligned source entries are comparable with native target entries. Groups #4, #6, and #8 originate from METR-LA but have alignment confidences in the same range as PEMS-BAY groups. In particular, group #8 supplies a sharp congestion-onset pattern that is difficult to obtain from only three days of target data.

### B.2 Case Study

The case in Fig. 5 comes from node 259 in PEMS-BAY under the M-P setting. The coarse temporal search returns three entries whose histories are similar to the target history, but their futures follow a sustained fall, a rise followed by a decline, and a drop followed by recovery. This divergence shows why temporal similarity alone cannot determine a reliable analogy. The Retrieval Agent therefore reranks the candidates using temporal semantics and aligned spatial embeddings, and the perspective weights in  $\mathcal{M}_{RT}$  determine the selected augmented entry  $b_{\text{aug}}$ .

Guided by  $b_{\text{aug}}$ , the Prediction Agent performs analogical forecasting over the query and the stored reference future. The target history first rises and then remains in a high-speed regime, while the ground truth contains an abrupt decrease followed by recovery.

Without retrieved local knowledge, w/o RAG Retrieval extrapolates a persistent decline and accumulates large long-horizon errors. R<sup>2</sup>-Transfer instead predicts a milder decline followed by stabilization. Although it does not reproduce the abrupt decrease exactly, it avoids the strong downward bias of the ablation and reduces MAE from 2.48 to 1.38.

This case illustrates that source-city experience is retained as explicit local entries and transferred according to the current query rather than through a global carrier. The explicit path from candidate retrieval to analogical prediction also makes the Retrieval Agent and Prediction Agent individually inspectable.

### B.3 Explainability

R<sup>2</sup>-Transfer produces an explicit reasoning trace for every prediction. The trace shows how the selected augmented entry supports the forecast and gives the Reflection Agent evidence for failure attribution. To assess retrieval quality, the Prediction Agent also receives lightweight metadata from the two unselected perspective nominees, including their temporal context, spatial category, and nomination perspective, but not their time series.

The representative M-P trace below demonstrates three operations. First, the Prediction Agent evaluates retrieval quality from temporal and spatial perspectives: the five-hour offset produces a poor temporal match, while the spatial match remains reliable. Second, it resolves the weak upward movement in the target history against a stable retrieved future and discounts the local slope as noise. Third, it checks whether the error patterns in  $\mathcal{M}_P$  match the current weekday, time range, and trend. Because no pattern applies, the agent avoids unnecessary compensation. Recording each judgment gives the Reflection Agent a complete evidence path for later attribution.

#### LLM Reasoning Output

##### Reasoning

##### Retrieval Quality Assessment

**Temporal Match:** Poor. No exact temporal match. Target is Sunday 00:00 (early morning weekend). Retrieved case is at 05:00 Sunday.

**Spatial Match:** Good. Retrieved case matches the same function (evening\_peak\_dominant).

**Spatial Alignment Reliability:** Reliable. Spatial semantics can be used as validation.

##### RAG Quality Assessment

**Diversity:** Low. All cases are evening\_peak\_dominant hubs. Temporal diversity is limited; no case matches the target's early morning weekend slot.

**Concern:** The lack of a temporally matched case at 00:00 weekend is the main weakness. Case 2 (Sunday 05:00) is the closest temporal match but still 5 hours off.

##### Inference Chain

**Baseline Analysis:** Target is Sunday 00:00 at an evening\_peak\_dominant node. Seasonal baseline for early morning weekend in such nodes is typically a stable, slightly declining or flat pattern as traffic is at its lowest point of the day. The historical sequence shows speeds around 64–65.6, with a very slight upward drift in the last few steps. The baseline expectation is for speeds to remain in the 64–66 range with minimal variation, possibly a very slight decline as the pre-dawn trough approaches.

**Trend Adjustment:** The local trend is rising direction, linear shape, but the rise is extremely subtle. Given the early morning hour and weekend context, this slight rise is likely noise or a temporary fluctuation, not a sustained upward trend. The temporal match shows a very stable pattern with a slight rise followed by a slight decline, with future speeds also stable around 67. This supports a stable-to-slightly-declining expectation. No pre-peak dip pattern is relevant here as it is not afternoon. The rising trend should be discounted as insignificant.

**Spatial Check:** Spatial alignment is reliable. The retrieved case is spatially similar (hub, evening\_peak\_dominant). The spatial semantics validate that early

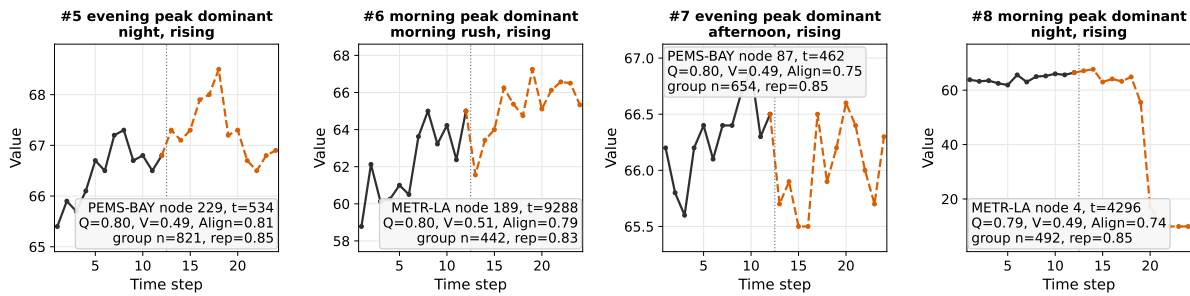


Figure 8: Representative entries of knowledge-base groups 5–8 under the M-P setting.

morning weekend at an evening\_peak\_dominant hub should be a quiet, stable period.

**Error Compensation:** No error patterns are triggered. The target is at 00:00 Sunday (early morning weekend), not at 02:00–03:00 weekday (Pattern 1,2), not at midday weekend (Pattern 3), not at morning rush (Pattern 4,5), not at weekend afternoon with subtle rise (Pattern 6), not at 05:00 weekday (Pattern 7), and not at weekday midday (Pattern 8). The current reasoning strategy about pre-peak dip is not applicable as this is not afternoon. The stable pattern with minimal noise suggests no compensation is needed.

**Confidence Rationale:** The target’s historical sequence is very stable with low noise, and the early morning weekend period is typically predictable. The temporal match confirms a stable pattern. Spatial alignment is reliable. No error patterns are triggered. The main uncertainty is the exact speed level, but the pattern (stable, slight decline) is clear. Prediction will center around 65 with minor fluctuations.

#### Prediction

[65.2, 65.0, 64.8, 64.7, 64.9, 65.1, 65.0, 64.8, 64.6, 64.7, 64.9, 65.0]

## B.4 Reflection Process Analysis

The Reflection Agent follows a closed chain of trigger, diagnosis, bounded revision, and keep-or-rollback verification. Table 3 traces this chain. The low trigger rates show that reflection responds to regime-inconsistent errors rather than every large absolute error. More than 90% of triggered reflections pass bounded-update validation, and more than 81% of applied revisions are retained. Most revisions target prediction memory  $M_P$  or the prediction prompt, whereas few target knowledge-base memory. This distribution suggests that the admission policy already keeps stored entries informative and nonredundant.

Table 5 evaluates the effect of applied revisions against both the triggering error and the pre-update EMA baseline. Retained revisions recover approximately 63% of the triggering error on M-P and N-D and bring window MAE to within 0.6 of the normal error level. Up to 46.61% of retained windows return below the EMA baseline despite following an anomalous regime. In contrast, revisions later rolled back would have increased window error beyond the trigger and more than 5.0 above the EMA baseline; none outperforms the baseline. This separation indicates that the keep-or-rollback rule reliably distinguishes beneficial updates from harmful ones.

## B.5 Efficiency Comparison

Table 4 compares four reasoning LLM backbones. GPT o1 achieves the best result in seven of eight evaluations and is particularly strong at long horizons, but it has the highest estimated cost at

\$92.75 per 1,000 queries. Claude Sonnet 4 remains close to GPT o1 while reducing estimated cost by 77.2%, although its average latency increases to 43.7 seconds. DeepSeek-R1 reduces cost further to \$3.39 per 1,000 queries, with moderate accuracy degradation and the highest latency. Gemini 2.0 Flash Experimental is the fastest and costs only \$1.47 per 1,000 queries, but produces larger errors. The comparison shows that the same R<sup>2</sup>-Transfer workflow can prioritize accuracy, accuracy-cost balance, monetary efficiency, or latency by changing the reasoning backbone.

## C Additional Related Work

### C.1 LLMs for Spatio-Temporal Forecasting

LLM-based time-series forecasting methods can be broadly divided into three groups. The first group represents numerical sequences in a form that language models can process directly. LLTime serializes numerical values into strings and performs forecasting through next-token prediction [4]. The second group adapts pretrained language models to temporal data. GPT4TS applies pretrained language models to general time-series analysis [48]. Time-LLM reprograms temporal inputs into representations compatible with a frozen language model [11]. TEST aligns time-series embeddings with textual prototypes [32]. LLM4TS adapts pretrained language models to data-efficient forecasting [2]. S<sup>2</sup>IP-LLM learns prompts from the semantic space of a language model [27]. aLLM4TS adapts language models through multi-patch prediction [1]. LangTime combines language-guided temporal representations with reinforcement learning-based fine-tuning [26]. The third group incorporates spatial information or urban knowledge into language-based forecasting. UrbanGPT integrates spatio-temporal representations with instruction tuning [18]. UrbanLLM applies LLM reasoning to urban activity planning and management [8]. FSTLLM uses the contextual knowledge of LLMs for few-shot spatio-temporal forecasting [9]. ST-LINK introduces spatially enhanced attention and memory retrieval for traffic forecasting [7]. Time-VLM combines temporal, textual, and visual information through a pretrained vision-language model [47].

Retrieval-augmented forecasting introduces external historical knowledge into the prediction process. RAG first establishes a general framework for combining retrieval and generation [15]. TimeRAG retrieves similar historical sequences to support LLM-based forecasting [39]. Retrieval-augmented diffusion models use related time series as references during denoising [20]. TimeRAF

learns a retriever for selecting information from a time-series knowledge base [45]. TS-RAG retrieves historical sequences for zero-shot forecasting [25]. RAFT directly combines the histories and futures of retrieved training samples with the current input [5]. Recent retrieval-augmented time-series models further study retrieval for time-series foundation models [34]. XRAG4TS extends retrieval-augmented forecasting to cross-city time-series prediction [10].

Agent-oriented LLM research provides mechanisms for iterative reasoning and feedback. ReAct interleaves reasoning with actions [41]. Self-Refine revises model outputs through self-generated feedback [24]. Reflexion stores verbal feedback in memory to guide

later decisions [31]. NewsForecast introduces reflection into event-aware time-series forecasting [36]. However, existing LLM-based and retrieval-augmented forecasting methods usually execute knowledge transfer and prediction through a fixed or monolithic pipeline. Knowledge base construction, knowledge retrieval, and prediction are not independently inspectable and revisable. A large prediction error therefore cannot be attributed to missing knowledge, irrelevant knowledge retrieval, or the misuse of relevant knowledge. The prediction loss only reflects the magnitude of the error. The prediction loss does not identify the step that caused the error. Correcting a failure often requires redesigning the prompt, fine-tuning the model, or retraining the entire system.